Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A reaction mixture containing $\mathrm{H_2}$, $\mathrm{N_2}$ and $\mathrm{NH_3}$ has partial pressures $2\ \mathrm{atm}, 1\ \mathrm{atm}$ and 3 atm respectively at $725\ \mathrm{K}$. If the value of $\mathrm{K_p}$ for the reaction, $\mathrm{N_{2(\ g)} + 3H_{2(\ g)} \rightleftharpoons 2NH_{3(\ g)}}$ is $4.28 \times 10^{-5}\ \mathrm{atm^{-2}}$ at $725\ \mathrm{K}$, in which direction the net reaction will go?

(A) Forward

(B) Backward

(C) No net reaction

(D) Direction of reaction cannot be predicted

Q.2

$\mathrm{K_1}$ and $\mathrm{K_2}$ are equilibrium constant for reactions (1) and (2)

$\mathrm{N_2(\ g) + O_2(\ g) \rightleftharpoons 2NO(g)} \quad \cdots (1)$

$\mathrm{NO(g) \rightleftharpoons \frac{1}{2}\ N_2(\ g) + \frac{1}{2} O_2(\ g)}$ ...(2)

Then,

(A) $\mathrm{K_1 = \left(\frac{1}{K_2}\right)^2}$

(B) $\mathrm{K_1 = K_2^2}$

(C) $\mathrm{K_1 = \frac{1}{K_2}}$

(D) $\mathrm{K_1 = (K_2)^0}$

Q.3

For the reaction

$\mathrm{2HI(g) \rightleftharpoons H_2(\ g) + I_2(\ g)}$

The degree of dissociation (α) of $\mathrm{HI(g)}$ is related to equilibrium constant, $\mathrm{K_p}$ by the expression

(A) $\mathrm{\frac{1+2\sqrt{K_p}}{2}}$

(B) $\mathrm{\sqrt{\frac{1+2\ K_p}{2}}}$

(C) $\mathrm{\sqrt{\frac{2\ K_p}{1+2\ K_p}}}$

(D) $\mathrm{\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}}$

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Q.4

In a reversible reaction, study of its mechanism says that both the forward and reverse reaction follows first order kinetics. If the half life of forward reaction $\left(t_{1/2}\right)_{f}$ is $400 \mathrm{sec}$ and that of reverse reaction $\left(t_{1/2}\right)_{r}$ is $250 \mathrm{sec}$. The equilibrium constant of the reaction is

(A) 1.6

(B) 0.433

(C) 0.625

(D) 1.109

Q.5

At temperature T, a compound $\mathrm{AB}_2(\mathrm{~g})$ dissociates according to the reaction $2 \mathrm{AB}_2(\mathrm{~g}) \rightleftharpoons 2 \mathrm{AB}(\mathrm{g})+\mathrm{B}_2(\mathrm{~g})$ with degree of dissociation α, which is small compared with unity. The expression for K_{p}, in terms of α and the total pressure, P_{T} is

(A) $\frac{\mathrm{P}_{\mathrm{T}} \alpha^3}{2}$

(B) $\frac{\mathrm{P}_{\mathrm{T}} \alpha^2}{3}$

(C) $\frac{\mathrm{P}_{\mathrm{T}} \alpha^3}{3}$

(D) $\frac{\mathrm{P}_{\mathrm{T}} \alpha^2}{2}$

Q.6

At a certain temperature the following equilibrium is established,

$\mathrm{CO}(\mathrm{g})+\mathrm{NO}_2(\mathrm{~g}) \rightleftharpoons \mathrm{CO}_2(\mathrm{~g})+\mathrm{NO}(\mathrm{g})$

One mole of each of the four gases is mixed in one litre container and the reaction is allowed to reach equilibrium state. When excess of baryta water is added to the equilibrium mixture, the weight of white precipitate obtained is $236.4 \mathrm{~g}$. The equilibrium constant, K_{C} of the reaction is

(A) 1.2

(B) 2.25

(C) 2.1

(D) 3.6

Q.7

For the reaction

$\mathrm{PCl}_{3(\mathrm{~g})}+\mathrm{Cl}_{2(\mathrm{~g})} \rightleftharpoons \mathrm{PCl}_{5(\mathrm{~g})}$

The value of K_{c} at $250^{\circ} \mathrm{C}$ is 26. The value of K_{p} at this temperature will be

(A) 0.61

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(B) 0.57

(C) 0.83

(D) 0.46

Q.8

In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be

(A) Reduced to half of its original value

(B) becomes (original)/4

(C) doubled

(D) constant

Q.9

For which of the following $\mathrm{K_p}$ may be equal to $0.5\ \mathrm{atm}$

(A) $2\mathrm{HI} \rightleftharpoons \mathrm{H_2} + \mathrm{I_2}$

(B) $\mathrm{PCl_5} \rightleftharpoons \mathrm{PCl_3} + \mathrm{Cl_2}$

(C) $\mathrm{N_2} + 3\mathrm{H_2} \rightleftharpoons 2\mathrm{NH_3}$

(D) $2\mathrm{NO_2} \rightleftharpoons \mathrm{N_2O_4}$

Q.10

The vapour density of undecomposed $\mathrm{N_2O_4}$ is 46. When heated, vapour density decreases to 25.4 due to its dissociation to $\mathrm{NO_2}$. The % dissociation of $\mathrm{N_2O_4}$ at the final temperature is

(A) 80

(B) 60

(C) 40

(D) 70

Q.11

The equilibrium constant for the reaction $\mathrm{W} + \mathrm{X} \rightleftharpoons \mathrm{Y} + \mathrm{Z}$ is 9. If one mole of each of W and X are mixed and there is no change in volume, the number of moles of Y formed is

(A) 0.10

(B) 0.50

(C) 0.75

(D) 0.54

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Q.12

When heated, ammonium carbamate decomposes as follows:
$\mathrm{NH_4COONH_2(\ s) \rightleftharpoons 2NH_3(\ g) + CO_2(\ g) \cdot At}$
a certain temperature, the equilibrium pressure of the system is $0.318\ \mathrm{atm}$. $\mathrm{K_p}$ for the reaction is:

(A) 0.128

(B) 0.426

(C) 4.76×10^{-3}

(D) None of these

Q.13

One mole of $\mathrm{SO_3}$ was placed in a one litre reaction flask at a given temperature when the reaction equilibrium was established in the reaction. $\mathrm{2SO_3 \rightleftharpoons 2SO_2 + O_2}$. The vessel was found
to contain 0.6 mole of $\mathrm{SO_2}$. The value of the equilibrium constant is

(A) 0.36

(B) 0.675

(C) 0.45

(D) 0.54

Q.14

For the equilibrium system

$$\mathrm{2HX(g) \rightleftharpoons H_2(\ g) + X_2(\ g)}$$

the equilibrium constant is 1.0×10^{-5}. What is the concentration of HX if the equilibrium concentration of $\mathrm{H_2}$ and $\mathrm{X_2}$ are $1.2 \times 10^{-3}\mathrm{M}$, and $1.2 \times 10^{-4}\mathrm{M}$ respectively.

(A) $12 \times 10^{-4}\mathrm{M}$

(B) $12 \times 10^{-3}\mathrm{M}$

(C) $12 \times 10^{-2}\mathrm{M}$

(D) $12 \times 10^{-1}\mathrm{M}$

Q.15

If $\mathrm{K_1}$ and $\mathrm{K_2}$ are respective equilibrium constants for the two reactions
$\mathrm{XeF_6(\ g) + H_2O(g) \rightleftharpoons XeOF_4(\ g) + 2HF(g)}$
$\mathrm{XeO_4(\ g) + XeF_6(\ g) \rightleftharpoons XeOF_4(\ g) + XeO_3\ F_2(\ g)}$
the equilibrium constant for the reaction $\mathrm{XeO_4(\ g) + 2HF(g) \rightleftharpoons XeO_3\ F_2(\ g) + H_2O(g)}$ will be

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(A) $\frac{K_1}{K_2^2}$

(B) $K_1 \cdot K_2$

(C) $\frac{K_1}{K_2}$

(D) $\frac{K_2}{K_1}$

Q.16

If 1.0 mole of I_2 is introduced into 1.0 litre flask at $1000\ K$, at quilibrium $\left(K_c = 10^{-6}\right)$, which one is correct

(A) $[I_2(g)] > [I^-(g)]$

(B) $[I_2(g)] < [I^-(g)]$

(C) $[I_2(g)] = [I^-(g)]$

(D) $[I_2(g)] = \frac{1}{2}[I^-(g)]$

Q.17

In a reaction $A + B \rightleftharpoons C + D$, the initial concentrations, of A and B were $0.9\ mol.dm^{-3}$ each. At equilibrium the concentration of D was found to be $0.6\ moldm^{-3}$. What is the value of equilibrium constant for the reaction

(A) 8

(B) 4

(C) 9

(D) 3

Q.18

The partial pressure of $CH_3OH(g), CO(g)$ and $H_2(g)$ in equilibrium mixture for the reaction, $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ are $2.0, 1.0$ and 0.1 atm respectively at $427^\circ C$. The value of K_p for the decomposition of CH_3OH to CO and H_2 is

(A) $10^2\ atm$

(B) $2 \times 10^2\ atm^{-1}$

(C) $50\ atm^2$

(D) $5 \times 10^{-3}\ atm^2$

Q.19

3.2 moles of hydrogen iodide were heated in a sealed bulb at $444^\circ C$ till the equilibrium state was reached. Its degree of dissociation at this temperature was found to be 22% The number of moles of hydrogen iodide

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present at equilibrium
are

(A) 2.496

(B) 1.87

(C) 2

(D) 4

Q.20

K_c for $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 0.04 at $250^\circ C$. How many moles of PCl_5 must be added to a $3\ L$ flask to obtain aCl_2 concentration of $0.15M$

(A) 4.2 moles

(B) 2.1 moles

(C) 5.5 moles

(D) 6.3 moles

Answers & Solutions

Q.1 Answer:

Backward

Solution:

$$Q_p = \frac{\left(P_{NH_3}\right)^2}{P_{N_2} \times \left(P_{H_2}\right)^3} = \frac{(3)^2}{(1)(2)^3} = \frac{9}{8}\ atm^{-2} \quad = 1.125\ atm^{-2}$$

Since value of Q_p is larger than K_p $\left(4.28 \times 10^{-5}\ atm^{-2}\right)$, it indicates net reaction will proceed in backward direction.

Q.2 Answer:

$$K_1 = \left(\frac{1}{K_2}\right)^2$$

Solution:

$$N_2 + O_2 \rightleftharpoons 2NO \quad \ldots K_1$$
$$2NO \rightleftharpoons N_2 + O_2 \quad \ldots \frac{1}{K_1}$$
$$NO \rightleftharpoons \rightleftharpoons \frac{1}{2} N_2 + O_2 \quad \ldots K_2 = \left(\frac{1}{K_1}\right)^{1/2}$$
$$K_2 = \left(\frac{1}{K_1}\right)^{1/2} \text{ or } K_2^2 = \frac{1}{K_1}$$
$$K_1 = \frac{1}{(K_2)^2}$$

Q.3 Answer:

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$\frac{2\sqrt{\mathrm{K_p}}}{1+2\sqrt{\mathrm{K_p}}}$

Solution:

$2\mathrm{HI(g)} \rightleftharpoons \mathrm{H_2(\ g)} + \mathrm{I_2(\ g)}$

$1-\alpha \qquad \frac{\alpha}{2} \qquad \frac{\alpha}{2}$

$K_p = \frac{\left(\frac{\alpha}{2} P_T\right)^2}{(1-\alpha)^2 P_T^2}$

$\frac{\alpha}{1-\alpha} = 2\sqrt{K_\mathrm{p}}$

$\alpha = \frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

Q.4 Answer:

0.625

Solution:

$K_f = \frac{0.693}{400}\ \mathrm{sec}^{-1}; K_r = \frac{0.693}{250}\ \mathrm{sec}^{-1}$

$K = \frac{k_f}{k_r} = \frac{250}{400} = 0.625$

Q.5 Answer:

$\frac{\mathrm{P_T}\alpha^3}{2}$

Solution:

For the given equilibrium, the equilibrium concentration are

$2\mathrm{AB_2(\ g)} \rightleftharpoons 2\mathrm{AB(g)} + \mathrm{B_2(\ g)}$

Equilib. conc. $c(1-\alpha) \quad c\alpha \quad \frac{Cd}{2}$

$\therefore \mathrm{K_P} = \frac{\left(\mathrm{P_{B_2}}\right)\left(\mathrm{P_{AB}}\right)^2}{\left(\mathrm{P_{AB_2}}\right)^2} = \frac{\frac{c\alpha}{2} \times (c\alpha)^2 \times \mathrm{P_T}}{[\mathrm{c}(1-\alpha)]^2\left[\mathrm{c}\left(1+\frac{\alpha}{2}\right)\right]}; \mathrm{K_P} = \frac{\alpha^3 \times \mathrm{P_T}}{2(1-\alpha)^2\left(1+\frac{\alpha}{2}\right)}$

Since, α **is small compared to unity,**

so $1-\alpha \simeq 1$ **and** $1+\frac{\alpha}{2} \simeq 1.$

$\mathrm{K_P} = \frac{\alpha^3 \times \mathrm{P_T}}{2}$

Q.6 Answer:

2.25

Solution:

$\mathrm{CO(g)} + \mathrm{NO_2(\ g)} \rightleftharpoons \mathrm{CO_2(\ g)} + \mathrm{NO(g)}$

Moles initially $1 \quad 1$

Moles at equib. $1-x 1-x \quad 1+x \quad 1+x$

$\mathrm{CO_2} + \mathrm{Ba(OH)_2} \longrightarrow \mathrm{BaCO_3} + \mathrm{H_2O}$

Moles of $\mathrm{BaCO_3} = \frac{236.4}{197} = 1.2$

Moles of $\mathrm{CO_2}$ **at equilibrium** $= 1.2$ **or,**

$1 + \mathrm{x} = 1.2; \mathrm{x} = 0.2 \quad \therefore \quad \mathrm{K_C} = \left(\frac{1+\mathrm{x}}{1-\mathrm{x}}\right)^2 = \left(\frac{1.2}{0.8}\right)^2 = 2.25$

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Q.7 Answer:

0.61

Solution:

$K_p = K_c(RT)^{\Delta n} = 26 \times (0.082 \times 523)^{-1} = 0.61$

Q.8 Answer:

constant

Solution:

K_c or K_p do not depend on concentration, but only on temperature

Q.9 Answer:

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Solution:

For $K_p = 0.5$ atm

$\Delta n = 1$ **(since the unit is atm)**

and $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$\Delta n = 1$

Q.10 Answer:

80

Solution:

$N_2O_4 \rightleftharpoons 2NO_2$

1 0 at initial

$1-\alpha \quad 2\alpha$ **at equilibrium**

$\therefore \frac{V \cdot D_{initial}}{V \cdot D_{final}} = \frac{n_{final}}{n_{initial}}$

$\frac{46}{25.4} = \frac{1+\alpha}{1}$

$1.8 = 1 + \alpha \Rightarrow \alpha = 0.8$ **or** 80%

Q.11 Answer:

0.75

Solution:

	W	+	X	$\rightleftharpoons$	Y	+	Z
Initial	1		1		0		0
At eqn.	$1-\alpha$		$1-\alpha$		α		α

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$$K_{eq} = 9 = \frac{\alpha^2}{(1-\alpha)^2}$$
$$\therefore \alpha = 0.75$$
$$\text{Moles of Y} = 075$$

Q.12 Answer:

4.76×10^{-3}

Solution:

$$P_{\text{total}} = 3P$$
$$\Rightarrow \quad P = \frac{0.318}{3} = 0.106$$
$$\therefore \quad K_p = 4P^3 = 4.76 \times 10^{-3}$$

Q.13 Answer:

0.675

Solution:

	$2SO_3$	$\rightleftharpoons$	$2SO_2$	+	O_2
Initial	1		0		0
At equilibrium	$1-2\alpha$		2α		α

$\therefore 2\alpha = 0.6, \quad \therefore \alpha = 0.3$
$[SO_3] = 1 - 2\alpha = 1 - 0.6 = 0.4$
$[SO_2] = 2\alpha = 0.6$
$[O_2] = \alpha = 0.3$
$K_{eq} = \frac{\alpha \times (2\alpha)^2}{(1-2\alpha)^2} = \frac{0.3 \times 0.6 \times 0.6}{0.4 \times 0.4} = 0.675$

Q.14 Answer:

$12 \times 10^{-2}\text{M}$

Solution:

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$$2HX(g) \rightleftharpoons H_2(g) + X_2(g)$$

At eqm.	1.2×10^{-3}M	1.2×10^{-4}M

At eqm.

$K_{eq} = \frac{[H_2][X_2]}{[HX]^2}$

$10^{-5} = \frac{1.2\times10^{-3}\times1.2\times10^{-4}}{[HX]^2}$

$[HX] = \sqrt{\frac{1.2\times1.2\times10^{-7}}{10^{-5}}}$

$= 1.2 \times 10^{-1}$

$= 12 \times 10^{-2}$M

Q.15 Answer:

$\frac{K_2}{K_1}$

Solution:

$K = K_2 \times \frac{1}{K_1} = \frac{K_2}{K_1}$

Q.16 Answer:

$[I_2(g)] > [I^-(g)]$

Solution:

$I_2 \rightleftharpoons 2I^-$

$1 - x \quad 2x$

$K_c = \frac{(2x)^2}{(1-x)} = 10^{-6}$

Soln. shows that $(1 - x) > 2x$

$\therefore [I_2(g)] > [I^-(g)]$

Q.17 Answer:

4

Solution:

$$A + B \rightleftharpoons C + D$$

Initial	0.9	0.9	0	0
At eqm.	0.3	0.3	0.6	0.6

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$K_c = \frac{0.6 \times 0.6}{0.3 \times 0.3} = 4$

Q.18 Answer:

$5 \times 10^{-3}\ \text{atm}^2$

Solution:

$K_p = \frac{p_{CH_3OH}}{p_{CO} \times p_{H_2}} = \frac{2}{1 \times (0.1)^2} = 200;$

For reverse reaction

$\frac{1}{K_p} = \frac{1}{200} = 5 \times 10^{-3}\ \text{atm}^2$

Q.19 Answer:

2.496

Solution:

$2HI \rightleftharpoons H_2 + I_2$**.It is** 22% **decomposed**

$\therefore \frac{3.20 \times 22}{100} = 0.704$

(3.2-0.704) is equal to HI present at equilibrium which is $= 2.496$

Q.20 Answer:

2.1 moles

Solution:

At equilibrium the moles of Cl_2 **must be**

$= 0.15 \times 3 = 0.45$

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Eqm. Conc. $\frac{x-0.45}{3} \quad \frac{0.45}{3} \quad \frac{0.45}{3}$

$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$

$\therefore 0.04 = \frac{0.15 \times 0.15}{(x-0.45)/3}$

$\therefore x = 2.1$ **moles**

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